Jacob Berger

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Education

Oregon State University — Corvallis, OR

Ph.D. & M.S. in Artificial Intelligence Expected June 2030

Advisor: Dr. Lizhong Chen, STAR Lab. Coursework: Deep Learning, Parallel Programming, Operating Systems II, Numerical ODEs/PDEs, Math of Data Science, Analysis of Algorithms, Machine Learning.

Honors B.S. in Applied Computer Science & Computational Mathematics Sept 2022 – June 2026

Dual degree; Summa Cum Laude (both degrees). AI concentration.
Accelerated Master's Program (graduate coursework as undergraduate).

Research Experience

Honors Thesis: Semantic Constraints in NLP June 2025 – June 2026

Researcher (Advisor: Dr. Lizhong Chen, STAR Lab)

- Developed Parsimonious Decoding and Vocabulary Efficiency Score (VES) for Natural Semantic Metalanguage; evaluated five LLMs, raising base-model semantic legality from 8.1–10.3% to 52.0–78.0%. First-author paper in preparation.

Experience

Graduate Research Assistant June 2026 – Present

Oregon State University, STAR Lab — Corvallis, OR

- Developing ML preconditioner selection and correction for parametric PDE solvers (transport) with CARRE and LLNL.
- Continuing honors-thesis research on LLM constrained decoding.

Technical Contributor & Cofounder March 2023 – Present

Lexel Synergetics — Portland, OR

- Led the technical proposal that secured a $305,000 NSF Phase I SBIR grant for NLP-based language model development.
- Architected end-to-end data pipelines for training and fine-tuning specialized LLMs (LoRA, custom loss functions, inference-time logit modulation) and deployed training and serving infrastructure on AWS, GCP, and Vercel.

Head Mechanic Dec 2022 – June 2026

MU Lanes and Games — Corvallis, OR

- Maintained and diagnosed electromechanical lane systems; managed several large center-level upgrades.

Projects

Senior Capstone: AI for Home Automation Sept 2025 – June 2026

- Integrated LLMs with Home Assistant for natural-language automation. Presented at OSU Engineering Research Expo.

AuTuber — AI Stage Hand for Live Streaming | github.com/waiyanzt/AuTuber May 2026 (BeaverHacks)

- LLM-driven agent observing webcam, screen, and audio to autonomously control OBS and VTube Studio via WebSocket. Team of 4; PM and technical lead.

RelCube — Deep RL Rubik's Cube Solver | github.com/bergerjacob/RelCube Winter 2026

- Transformer-based solver (∼3.5M params) using structured piece representation. Trained with Approximate Value Iteration and Expanding Curriculum.

Skills

- **Languages:** Python, C++, Java, JavaScript, Bash, x86 Assembly, MATLAB
- **Machine Learning:** PyTorch, HuggingFace Transformers, LLM Fine-tuning (LoRA), RAG Pipelines, Reinforcement Learning
- **Systems & Cloud:** Linux (Debian), Proxmox VE, Virtualization (LXC/KVM), AWS (EC2/S3), GCP, Git, Vercel
- **Parallel/HPC:** MPI, OpenMP, CUDA, Slurm
- **Human Languages:** Fluent Spanish (9 years immersion)

Awards

- Putnam Math Competition: Top 3 finisher at OSU (2022 competition); top 29% nationally (2025 competition).
- Finley Academic Excellence Scholarship: $20,000 merit scholarship, 2022.

Extracurriculars

- **Competitive Math:** OSU Putnam and COMAP Math Modeling (implemented Cellular Automata for ecosystem resilience analysis).
- **Competitive Sports:** Founder and current president of OSU national collegiate bowling team (All-Star qualifier), managing a $25,000+ annual budget; Rose City Rowing Club leader (2-time national qualifier).